Experiment 10

Dimensionality Reduction: Principal Component Analysis (PCA)

Aim: To apply Principal Component Analysis (PCA) for dimensionality reduction by transforming high-dimensional data into a lower-dimensional space while retaining the maximum variance.

Theory:

Principal Component Analysis (PCA) is a dimensionality reduction technique used in unsupervised learning to transform a dataset with many variables into a smaller set that still contains most of the essential information. PCA helps eliminate less important or redundant features, as a result it leads to faster training and inference. It also makes data visualization easier, especially for high-dimensional datasets. High-dimensional data often includes correlated features, which increase computational complexity and make interpretation difficult. PCA simplifies this problem by transforming the original variables into a new set of uncorrelated variables called principal components, while retaining as much of the original information as possible (Géron, 2019; Jolliffe, 2002).Geometrically, PCA rotates the coordinate system of the dataset so that the new axes align with the directions of maximum variation in the data.

1. **Objective of PCA**

The main objective of PCA is to identify directions in the feature space along which the variance of the data is maximized. Features with higher variance carry more information about the structure of the dataset. Therefore, PCA identifies directions where the data varies the most and projects the data onto those directions. These directions represent the most significant patterns in the dataset. In high-dimensional datasets, many features may contain redundant information due to correlation. PCA identifies new axes called principal components, which represent directions of maximum variance in the data.

The dataset is initially represented in a high-dimensional space as shown in figure. PCA then determines new orthogonal axes that capture the largest spread of the data. The original data points are projected onto these new axes, resulting in a reduced-dimensional dataset that still preserves most of the important information (Murphy, 2012).

2. **Geometric Interpretation of PCA**

Principal Component Analysis (PCA) can also be understood from a geometric perspective. In a multidimensional dataset, each feature represents an axis in a coordinate system. The data points are distributed within this high-dimensional space, often showing correlations among features.

PCA transforms the coordinate system by rotating the original axes to align with the directions of maximum variance in the data. These new axes are called principal components, and they provide a more compact representation of the dataset.
The first principal component (PC1) represents the direction along which the variance of the data is maximized. The second principal component (PC2) is orthogonal to the first and captures the second-largest variance. Subsequent principal components follow the same principle, each accounting for progressively smaller amounts of variance.

By projecting the original data points onto these new axes, PCA transforms the dataset into a new coordinate system that preserves the most significant patterns while reducing dimensionality. This geometric transformation allows complex high-dimensional data to be represented using fewer dimensions while retaining most of the important information.
As illustrated in Fig. 1, PCA identifies the directions of maximum variance using eigenvectors derived from the covariance matrix and projects the data onto these principal components to obtain a lower-dimensional representation.

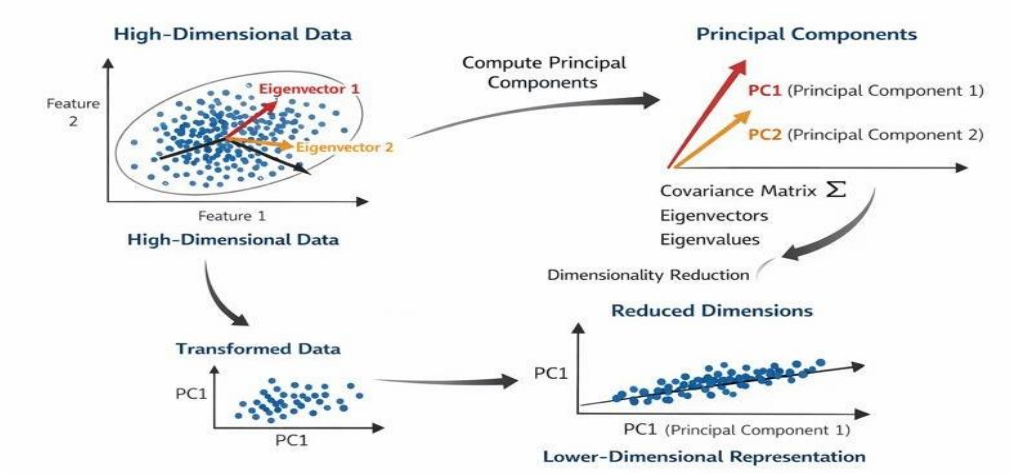


Fig. 1. Geometric interpretation of Principal Component Analysis (PCA) showing the transformation of high-dimensional data into principal components aligned with directions of maximum variance.

3. Mathematical Foundation

The mathematical foundation of PCA lies in linear algebra, specifically in eigenvectors and eigenvalues.
Eigenvectors define the directions of maximum variance in the data, while eigenvalues indicate the amount of variance explained by each direction. (Jolliffe, 2002).
For a square matrix A, an eigenvector v and its corresponding eigenvalue λ satisfy the equation:

$$Av = \lambda v$$

In PCA, eigenvectors and eigenvalues are derived from the covariance matrix of the dataset.The covariance matrix captures the relationships between all pairs of features and indicates how features vary together. PCA analyzes this matrix to identify directions where the data spreads the most.
Eigenvectors associated with larger eigenvalues correspond to more informative principal components because they capture more variance in the data. (Murphy, 2012 ;J. Brownlee, PCA for dimensionality reduction).

4. Dimensionality Reduction

By projecting the dataset onto a smaller number of important principal components, PCA reduces the number of features while keeping most of the useful information. The original high-dimensional data points are projected onto a lower-dimensional subspace formed by the selected principal components. Only the components that capture a large amount of variation in the data are selected.This process simplifies the dataset and helps reduce noise, making machine learning models easier to train and interpret (Géron, 2019; GeeksforGeeks).

5. Algorithm

The following steps describe the standard workflow used to perform Principal Component Analysis.

Step 1: Standardize the Data

Before applying PCA, the dataset is standardized so that each feature contributes equally to the analysis.

$$X_{standardized} = \frac{X - \mu}{\sigma}$$

Where,μ is the mean of the feature and σ is the standard deviation. After standardization, all features have a mean of 0 and a standard deviation of 1.

Step 2: Compute the Covariance Matrix

After standardization, the covariance matrix is computed to measure how features vary with respect to each other.

$$Cov(X) = \frac{1}{n-1} X^T X$$

The resulting covariance matrix has a size of d × d, where d represents the number of features. Each element [i, j] indicates the covariance between feature i and feature j.

Step 3: Calculate Eigenvalues and Eigenvectors

The eigenvalues and eigenvectors of the covariance matrix are computed to identify the principal directions of the data.

$$Av = \lambda v$$

Where,λ represents the eigenvalue (amount of variance captured) and v represents the eigenvector (direction of the new axis).

Step 4: Sort Eigenvectors by Eigenvalues

The eigenvectors are sorted in descending order based on their corresponding eigenvalues. The eigenvector with the largest eigenvalue represents the direction of maximum variance, followed by the next largest, which is orthogonal to the first.

Step 5: Compute Explained Variance Ratio

$$Explained\ Variance\ Ratio = \frac{\lambda_i}{\sum \lambda}$$

This ratio indicates the proportion of total variance explained by each principal component. The explained variance ratios are often visualized using a scree plot to determine how many principal components should be retained.

Step 6: Select Number of Principal Components (k)

The number of principal components k is selected based on the cumulative explained variance. A common approach is to select k such that the cumulative variance exceeds a predefined threshold (e.g., 95%).

Step 7: Construct Projection Matrix

A projection matrix W is created using the top k eigenvectors as its columns.

$$W = [v_1, v_2, \ldots, v_k]$$

Step 8: Transform the Dataset

$$X_{reduced} = X_{standardized} W$$

The transformed dataset now has k dimensions instead of the original d dimensions while preserving the most important variance in the data.

6. Merits of PCA

- Makes large and complex datasets easier to analyze by reducing feature dimensionality.
- Removes redundancy caused by correlated variables.
- Improves computational efficiency for machine learning algorithms.

- Helps visualize high-dimensional data in two or three dimensions.

7. Demerits of Using PCA

- The new features created by PCA (principal components) are linear combinations of original features and are difficult to interpret.
- Some useful information may be lost during dimensionality reduction.
- PCA assumes linear relationships between variables and may not work well for datasets with complex non-linear patterns.
- PCA is sensitive to outliers, which can influence the direction of principal components.
- PCA is sensitive to the scaling of the data, so proper standardization is required (Jolliffe, 2002).

Procedure:

The objective of this experiment is to determine whether a breast tumor is malignant or benign by applying Principal Component Analysis (PCA) to the Wisconsin Breast Cancer dataset. By transforming the original 30 features into a smaller set of principal components, the experiment aims to simplify the dataset, reduce redundancy, and improve visualization and model efficiency without significantly affecting classification performance.

Step 1: Import required libraries: pandas, numpy, matplotlib, seaborn, and required modules from scikit-learn.

Step 2: Load the dataset containing breast cancer diagnostic features and the diagnosis label. The dataset contains 569 samples and 32 columns. Out of these, one column is an ID, one is the target column called diagnosis (with values M for malignant and B for benign), and the remaining 30 columns are numerical features describing cell characteristics used for prediction of diagnosis.

Step 3: Perform exploratory data analysis (EDA).

Step 4: Check class distribution of the diagnosis variable to observe the number of benign and malignant samples.

Step 5: Define feature variables and target variable:

- Features (X): All diagnostic attributes
- Target (y): Diagnosis label

Step 6: Visualize feature correlations using a heatmap before applying PCA.

Step 7: Standardize the feature values using StandardScaler to achieve zero mean and unit variance.

Step 8: Split the dataset into training and testing sets using stratified sampling.

Step 9: Train a Logistic Regression model using the original (non-PCA) feature set.

Step 10: Evaluate the model performance before PCA using classification accuracy.

Step 11: Apply Principal Component Analysis (PCA) while retaining 95% of the total variance.

Step 12: Observe the reduced dimensionality after applying PCA.

Step 13: Train a Logistic Regression model using PCA-transformed features.

Step 14: Evaluate the model performance after PCA using classification accuracy.

Step 15: Compare model accuracy before and after PCA using a bar plot.

Step 16: Apply PCA with a fixed number of components to compute feature loadings.

Step 17: Visualize feature contributions to principal components using a heatmap.

Step 18: Analyze explained variance ratio to understand the contribution of principal components.

PCA - Practical Implementation

Step 1: Importing Libraries

 # Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import ipywidgets as widgets
from IPython.display import display
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from ipywidgets import interact, IntSlider
print("Libraries Imported")
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Load the Breast cancer Wisconsin dataset

```
df = pd.read_csv('Breast_cancer_Wisconsin_data.csv')
print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

 # Display the first 5 rows of the dataset

```
df.head()
```

Output:

	id	radi..	text..	perime..	are..	smooth..	compact..	concav..	concave p..	symme..	....	textu..	perime..	are..	smoothn..	compact..	concav..	concave p..	symmet..	fractal_dim..	dia..
0	84..	17.99	10.38	122.80	100..	0.11840	0.27760	0.3001	0.14710	0.2419	...	17.33	184.60	201..	0.1622	0.6656	0.7119	0.2654	0.4601	0.11890	M
1	84..	20.57	17.77	132.90	132..	0.08474	0.07864	0.0869	0.07017	0.1812	...	23.41	158.80	195..	0.1238	0.1866	0.2416	0.1860	0.2750	0.08902	M 2
84..	19.69	21.25	130.00	120..	0.10960	0.15990	0.1974	0.12790	0.2069	...	25.53	152.50	170..	0.1444	0.4245	0.4504	0.2430	0.3613	0.08758	M 3	
84..	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	...	26.50	98.87	567.7	0.2098	0.8663	0.6869	0.2575	0.6638	0.17300	M 4	
84..	20.29	14.34	135.10	129..	0.10030	0.13280	0.1980	0.10430	0.1809	...	16.67	152.20	157..	0.1374	0.2050	0.4000	0.1625	0.2364	0.07678	M	

Output:

```
5 rows x 32 columns
```

Displays the last five rows of the dataset

```
df.tail()
```

Output:

	id	radi	text.	perime	are	smooth.	compact.	concav	concave p	symme.	..	textu.	perime.	are.	smoothn.	compact.	concav.	concave p.	symmet	fractal_dim.	dia
564	92.	21.56	22.39	142.00	147	0.11100	0.11590	0.2439	0.13890	0.1726	...	26.40	166.10	202.	0.14100	0.21130	0.4107	0.2216	0.2060	0.07115	M
565	92.	20.13	28.25	131.20	126	0.09780	0.10340	0.1440	0.09791	0.1752	...	38.25	155.00	173.	0.11660	0.19220	0.3215	0.1628	0.2572	0.06637	M
566	92.	16.60	28.08	108.30	858	0.08455	0.10230	0.0925	0.05302	0.1590	...	34.12	126.70	112.	0.11390	0.30940	0.3403	0.1418	0.2218	0.07820	M
567	92.	20.60	29.33	140.10	126	0.11780	0.27700	0.3514	0.15200	0.2397	...	39.42	184.60	182.	0.16500	0.86810	0.9387	0.2650	0.4087	0.12400	M
568	92.	7.76	24.54	47.92	181	0.05263	0.04362	0.0000	0.00000	0.1587	...	30.37	59.16	268.	0.08996	0.06444	0.0000	0.0000	0.2871	0.07039	B

Output:

```
5 rows x 32 columns
```

Check dataset shape

```
df.shape
```

Output:

```
(569, 32)
```

Displays summary of dataset, including column names, data types, and non-null counts.

```
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
```

Output:

#	Column	Non-Null Count	Dtype
0	id	569 non-null	int64
1	radius_mean	569 non-null	float64
2	texture_mean	569 non-null	float64
3	perimeter_mean	569 non-null	float64
4	area_mean	569 non-null	float64
5	smoothness_mean	569 non-null	float64
6	compactness_mean	569 non-null	float64
7	concavity_mean	569 non-null	float64
8	concave points_mean	569 non-null	float64
9	symmetry_mean	569 non-null	float64
10	fractal_dimension_mean	569 non-null	float64
11	radius_se	569 non-null	float64
12	texture_se	569 non-null	float64
13	perimeter_se	569 non-null	float64

14	area_se	569 non-null	float64

... 17 more rows (total: 32 rows)

Output:

```
dtypes: float64(30), int64(1), object(1)
memory usage: 142.4+ KB
```

Statistical summary

```
df.describe()
```

Output:

	id	radi	text	perim	area	smooth	compac	conca	concave	symme	..	radi	textu	perime	area	smooth	compact	concav	concave p	symme	fractal_di
co	5.69	569	569	569.0	569	569.00	569.00	569.0	569.00000	569.0	...	569	569.0	569.00	569	569.00	569.000	569.00	569.000000	569.0	569.000000
mea	3.03	14.1	19.2	91.96	654	0.09636	0.10434	0.088	0.048919	0.181	...	16.2	25.67	107.26	880	0.13236	0.254265	0.27218	0.114606	0.290	0.083946
std	1.25	3.52	4.30	24.29	351	0.01406	0.05281	0.079	0.038803	0.027	...	4.83	6.146	33.602	569	0.02283	0.157336	0.20862	0.065732	0.061	0.018061
min	8.67	6.98	9.71	43.79	143	0.05263	0.01938	0.000	0.000000	0.106	...	7.93	12.02	50.410	185	0.07117	0.027290	0.00000	0.000000	0.156	0.055040
25%	8.69	11.7	16.1	75.17	420	0.08637	0.06492	0.029	0.020310	0.161	...	13.0	21.08	84.110	515	0.11660	0.147200	0.11450	0.064930	0.250	0.071460
50%	9.06	13.3	18.8	86.24	551	0.09587	0.09263	0.061	0.033500	0.179	...	14.9	25.41	97.660	686	0.13130	0.211900	0.22670	0.099930	0.282	0.080040
75%	8.81	15.7	21.8	104.1	782	0.10530	0.13040	0.130	0.074000	0.195	...	18.7	29.72	125.40	1084	0.14600	0.339100	0.38290	0.161400	0.317	0.092080
max	9.11	28.1	39.2	188.5	2501	0.16340	0.34540	0.426	0.201200	0.304	...	36.0	49.54	251.20	4254	0.22260	1.058000	1.25200	0.291000	0.663	0.207500

Output:

```
8 rows x 31 columns
```

Counts the frequency of each unique category

```
df['diagnosis'].value_counts()
```

Output:

```
count
diagnosis
B          357
M          212
Name: count, dtype: int64
```

Step 4: Data Preprocessing

Separate features and target variable

```
X = df.drop('diagnosis', axis=1)
y = df['diagnosis']
print("Feature shape:", X.shape)
print("Target shape:", y.shape)
```

Output:

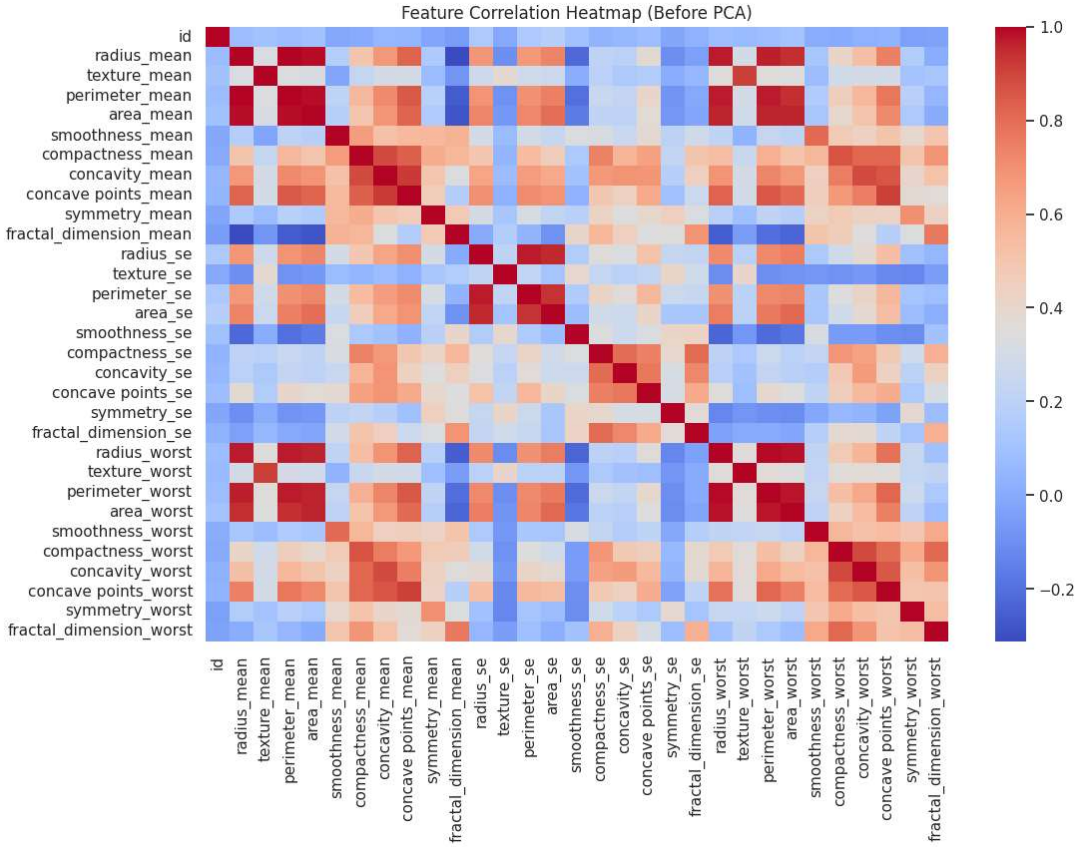
```
Feature shape: (569, 31)
Target shape: (569,)
```

Visualize correlations among features before applying PCA

```
plt.figure(figsize=(12,8))
sns.heatmap(X.corr(), cmap='coolwarm')
plt.title("Feature Correlation Heatmap (Before PCA)")
plt.show()
```

Output:

```
Feature Correlation Heatmap (Before PCA)
```



 # Standardize the features

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("Features standardized using StandardScaler")
```

Output:

```
Features standardized using StandardScaler
```

Step 5: Model Training

 # Data Splitting

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)
print("Data has been Splitted")
```

Output:
```
Data has been Splitted
```

 # Apply PCA, and visualize variance

```
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train)
X_test_pca = pca.transform(X_test)
print("PCA applied with 95% variance retained.")
```

Output:
```
PCA applied with 95% variance retained.
```

 # Model training using Logistic Regression

```
clf = LogisticRegression(max_iter=2000)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
acc_before = accuracy_score(y_test, y_pred)
clf_pca = LogisticRegression(max_iter=2000)
clf_pca.fit(X_train_pca, y_train)
y_pred_pca = clf_pca.predict(X_test_pca)
acc_after = accuracy_score(y_test, y_pred_pca)
print("Model Training completed")
```

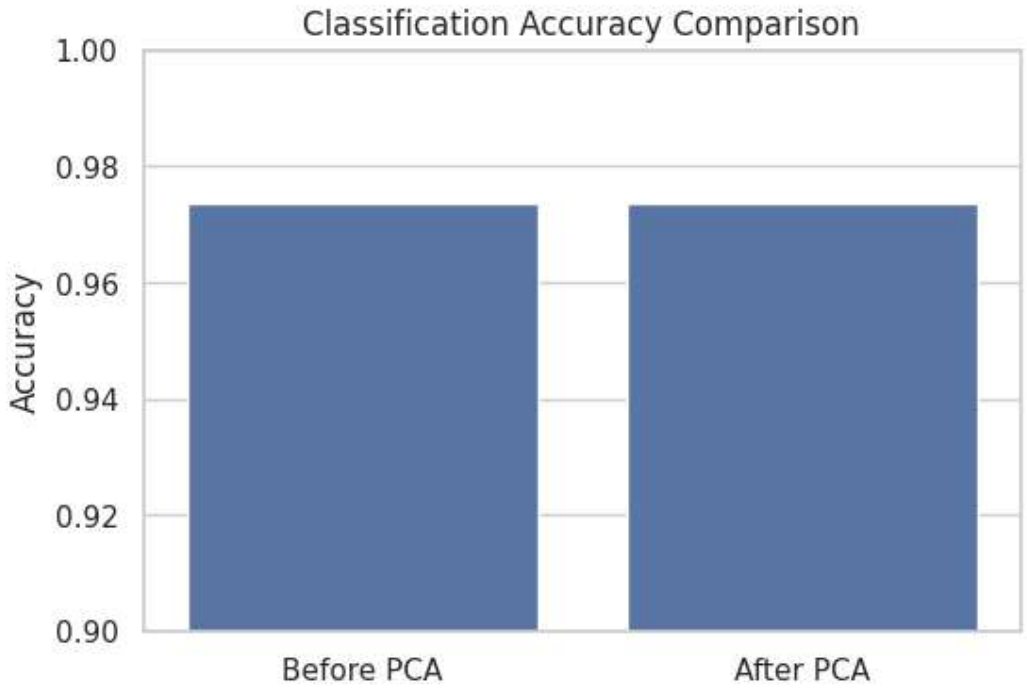
Output:
```
Model Training completed
```

Step 6: Model Evaluation

 # Compare classification accuracy before and after applying PCA

```
print("Accuracy BEFORE PCA:", round(acc_before*100, 2), "%")
pca = PCA(n_components=0.95)
X_pca = pca.fit_transform(X_scaled)
print("Reduced Dimensions:", X_pca.shape[1])
X_train_pca, X_test_pca, y_train, y_test = train_test_split(
    X_pca, y, test_size=0.2, random_state=42, stratify=y
)
print("Accuracy AFTER PCA:", round(acc_after*100, 2), "%")
plt.figure(figsize=(6,4))
sns.barplot(
    x=['Before PCA', 'After PCA'],
    y=[acc_before, acc_after]
)
plt.ylabel("Accuracy")
plt.title("Classification Accuracy Comparison")
plt.ylim(0.9, 1.0)
plt.show()
```

Output:
```
Accuracy BEFORE PCA: 97.37 %
Reduced Dimensions: 11
Accuracy AFTER PCA: 97.37 %
```



 # Apply PCA and compute feature loadings for each principal component

```
pca = PCA(n_components=10)
X_pca = pca.fit_transform(X_scaled)
loadings = pd.DataFrame( pca.components_.T, columns=[f"PC{i+1}" for i in range(10)],
    index=X.columns)
print("PCA feature loadings calculated for the top 10 principal components.")
```

Output:

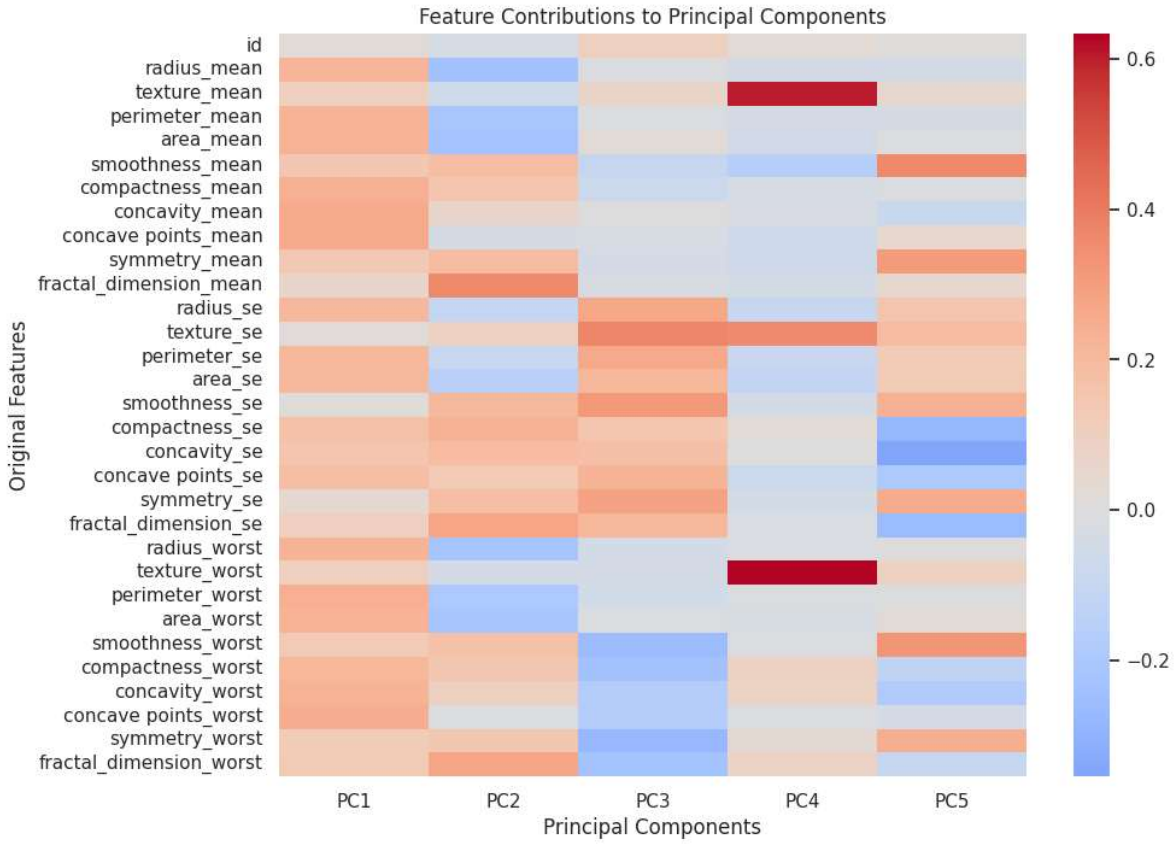
```
PCA feature loadings calculated for the top 10 principal components.
```

 # Visualize feature contributions to the first few principal components using a heatmap

```
plt.figure(figsize=(10,8))
sns.heatmap(loadings.iloc[:, :5], cmap="coolwarm",center=0)
plt.title("Feature Contributions to Principal Components")
plt.xlabel("Principal Components")
plt.ylabel("Original Features")
plt.show()
```

Output:

```
Feature Contributions to Principal Components
```



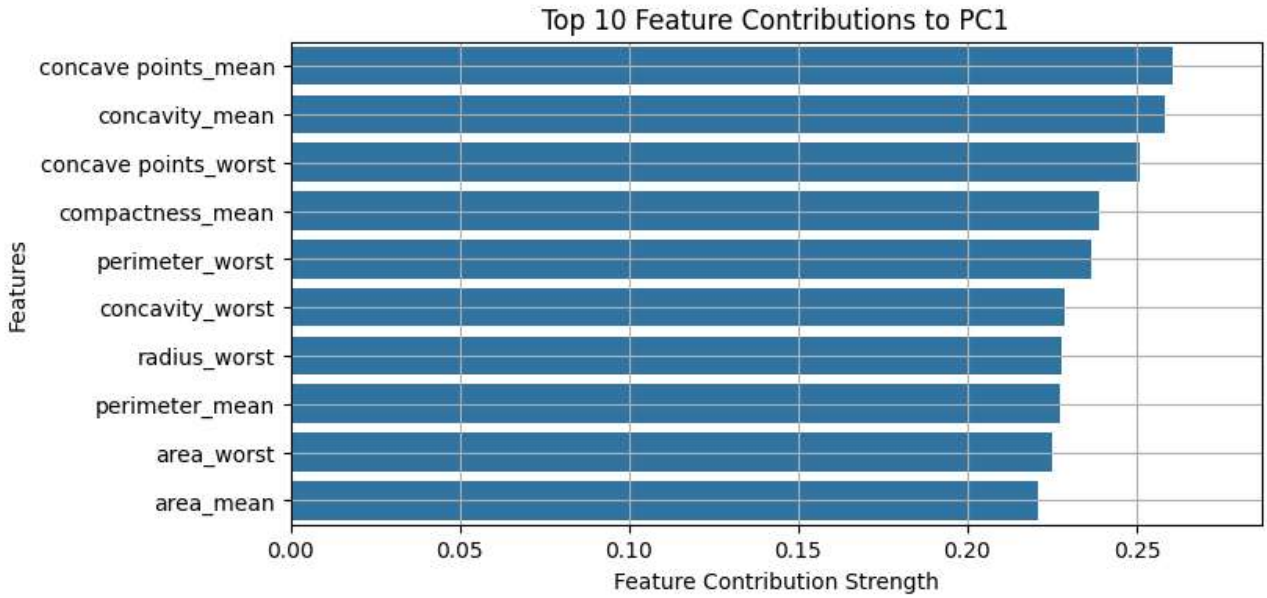
 # Interactively analyze and visualize top feature contributions for a selected principal component

```
def interactive_pca(pc_num=1):
    pc = f"PC{pc_num}"
    print(f"Principal Component: {pc}")
    print(f"Explained Variance Ratio: {pca.explained_variance_ratio_[pc_num-1]:.4f}")
    print(f"Cumulative Variance till {pc}:
    {pca.explained_variance_ratio_[:pc_num].sum():.4f}")
    top_features = loadings[pc].abs().sort_values(ascending=False).head(10)
    plt.figure(figsize=(8,4))
    sns.barplot( x=top_features.values, y=top_features.index)
    plt.xlim(0, top_features.max() * 1.1)
    plt.xlabel("Feature Contribution Strength")
    plt.ylabel("Features")
    plt.title(f"Top 10 Feature Contributions to {pc}")
    plt.grid(True)
    plt.show()

widgets.interact(interactive_pca,
    pc_num=widgets.IntSlider(min=1,max=10,step=1,value=1,description="PCA Component"));
```

Output:

```
Select any PCA component to visualize its feature contributions:
PCA Comp...
1
Principal Component: PC1
Explained Variance Ratio: 0.4286
Cumulative Variance till PC1: 0.4286
```



Interactive Features (Static View)

Interactive Feature: PCA Component Variance Viewer

In the simulation, a slider allows you to select PCA components 1-10 and view feature contributions. Below is the variance data for all components:

Component	Explained Variance Ratio	Cumulative Variance
PC1	0.4286	0.4286
PC2	0.1838	0.6124
PC3	0.0915	0.7039
PC4	0.0639	0.7678
PC5	0.0532	0.8210
PC6	0.0398	0.8608
PC7	0.0316	0.8924
PC8	0.0217	0.9140
PC9	0.0149	0.9289
PC10	0.0130	0.9419